

Pharmacy Stock Redistribution & Recommendation System

Product Name: PharmaShift – Expiry-Aware Pharmacy Network Optimizer

Project Parameter	Academic Specification
Project Title	Pharmacy Stock Redistribution & Recommendation System
Product Name	PharmaShift – Expiry-Aware Pharmacy Network Optimizer
Review Milestone	Project Review #3 (Evaluation Milestone)
Review #1 Score	34.3 / 35 marks (98% criteria met)
Review #2 Score	32.2 / 35 marks (92% criteria met)
Automated Tests	49 / 49 Passed across 16 categories (100% pass rate in 50.03s)
Production Build	Vite Production Bundle (2,355 modules, 0 errors)
ML Regressor	Random Forest (R ² : 0.7523, MAE: 1.2152, RMSE: 3.5170)
Simulation Validation	30 cycles: ■83.14L protected (+■7.44L over FIFO), -52.09% loss
Report Date	September 29, 2026

Student & Department Placeholders:

- Student Name(s): [Student Name(s) Placeholder]
- Register / Roll Number(s): [Register / Roll Number(s) Placeholder]
- Project Supervisor / Guide: [Project Supervisor Name Placeholder]
- Department of Computer Science & Engineering / Information Technology

2. Executive Summary & Review #2 Feedback Resolution

PharmaShift is an AI-assisted pharmacy inventory optimization platform designed to eliminate near-expiry medicine wastage and resolve localized stockouts across distributed pharmacy networks. Following Review #1 (34.3/35, 98%) and Review #2 (32.2/35, 92%), Review #3 comprehensively addresses all evaluator feedback recommendations.

Review #2 Evaluator Feedback	Review #3 Implementation Deliverable	Verification Evidence
1. Granular Technical Documentation on Unit Testing	Expanded test suite from 30 to 49 automated tests across 16 categories.	backend/tests/test_backend.py (49/49 passed in 50.03s)
2. Technical Documentation on Error Boundaries	Built React ErrorBoundary.jsx wrapping all 8 views with non-destructive state reset.	frontend/src/components/ErrorBoundary.jsx, App.jsx
3. Expanded Code Comments Explaining Logic	Added in-depth 'WHY' docstrings across clinical, mathematical, and security routines.	engine.py, feasibility.py, risk_scorer.py, ml_service.py
4. Document API Endpoints in README	Master API Directory (20 endpoints) with paths, methods, auth, roles, and status codes.	README.md Section 12, docs/api.md
5. Document Database Schema in README	Relational Schema (10 models) with constraints, foreign keys, and Mermaid ER diagram.	README.md Section 11, docs/database_schema.md
6. Strict Login-First Architecture	Removed auto-bypass; unauthenticated requests intercepted to Login; Quick-Fill panel added.	frontend/src/pages/Login.jsx, App.jsx

3. Core Implementations in Review #3 Category

- 3.1 Strict Login-First & Route Protection:** Unauthenticated users cannot view operational views. Direct URL access is intercepted by App.jsx, redirecting immediately to Login.jsx. Includes a Review Evaluation Quick-Fill panel with 1-click persona selection (Admin, Manager, Pharmacist 1, Pharmacist 2).
- 3.2 Role-Based Access Control (RBAC):** Passwords hashed with bcrypt (10 rounds, 72-byte safe truncation). Stateless HS256 JWT tokens. Strict branch scoping: a Pharmacist assigned to PHARM-001 is prohibited from accessing or approving transfers for PHARM-002 (HTTP 403 Forbidden).
- 3.3 React Error Boundaries & Fault Isolation:** ErrorBoundary.jsx encloses all 8 core views. Catches rendering exceptions, displays a controlled fallback card, and provides a 'Retry Section' non-destructive reset button that restores component state without user logout.
- 3.4 Backend Centralized Exception Shielding:** Global handlers for HTTPException, RequestValidationError (422), and unhandled 500 errors. Zero stack trace leakage to API consumers while maintaining structured server-side diagnostic logs.

4. Automated Unit Testing Results (49 / 49 Passed)

Command executed: `pytest -v backend/tests/test_backend.py`. **49 passed, 0 failed in 50.03 seconds (100% pass rate).**

Category	Count	Scope / Functionality Verified	Status
1. Authentication	4	Login success, invalid password (401), inactive user, me profile	PASSED
2. Authorization / RBAC	6	Branch scoping, cross-branch approval block (403), seed admin check	PASSED
3. Inventory	3	Batch listing, category filter, single batch lookup	PASSED
4. Pharmacy Network	2	18 branches listing, single branch details, 404 missing branch	PASSED
5. Demand & ML	4	Live RF inference, model metrics, 3-tier fallback chain, catalog	PASSED
6. Expiry-Risk Class.	4	DTE math, risk brackets, fixed demo reference date, date parser	PASSED
7. Safety-Stock	2	3-day safety buffer reservation, zero excess enforcement	PASSED
8. Recommender Engine	3	End-to-end generation, ML demand weighting, evidence bullets	PASSED
9. Destination Ranking	2	Haversine transit calculation, critical medication score bonus	PASSED
10. Feasibility Checks	7	Post-transit life >= 3 days, closed/maintenance branch, expired batch	PASSED
11. API Validation	2	Negative override rejection (422), missing rejection reason (422)	PASSED
12. Database Integrity	1	ORM foreign keys and cascade integrity traversal across models	PASSED
13. Audit Trail	4	Immutable log on approve, reject, override, and action filtering	PASSED
14. Analytics	2	Dashboard KPIs aggregation, 30-scenario simulation endpoint	PASSED
15. Edge Cases	1	Deterministic sandbox verifying all 8 failure prevention scenarios	PASSED
16. Error Handling	2	404 handling on invalid ID, subsystem health telemetry output	PASSED

5. Machine Learning & Empirical Simulation Validation

ML Demand Regressor: RandomForestRegressor (100 trees, depth 10) trained on 1,080 historical dispensing records:
MAE = 1.2152 units/day, RMSE = 3.5170 units/day, R² = 0.7523.

Empirical Monte Carlo Simulation (30 Cycles Benchmark vs Isolated FIFO):

Evaluation Metric	Baseline (Isolated FIFO)	Proposed Recommender	Net Gain / Impact
Average Value Protected	■75,70,344.23	■83,14,345.38	+■7,44,001.15 (+9.88%)
Value Lost to Expiration	■14,28,450.00	■6,84,448.85	-■7,44,001.15 (-52.09% loss reduction)
Clinician Acceptance Rate	N/A	93.03%	Empirical clinical concurrence
Average Transit Distance	N/A	10.27 km	Localized urban routing efficiency
Average Remaining Shelf Life	N/A	19.41 days	Generous safety margin post-transfer

6. 17-Step Live Evaluator Demonstration Script Summary

- **1. Unauthenticated** Interception: Access `http://localhost:5173/` -> verifies redirect to Login.
- **2. Quick-Fill Panel**: Select 'Admin' -> auto-populates `admin@pharmacy.io`.
- **3. Authentication**: Click 'Sign In' -> JWT token issued, redirect to Protected Dashboard.
- **4. Telemetry**: Verify health strip (Database Connected, ML Model $R^2=0.7523$, Ref Date 2026-08-14).
- **5. AI Action Center**: Review prioritized high-impact redistribution recommendations.
- **6. Inventory Monitor**: Filter by 'Critical' risk tier; observe 3-day safety stock reservation.
- **7. Explainability**: Inspect normalized 0-100 Recommendation Score and safety checklist.
- **8. Clinical Approval**: Approve standard transfer -> state transitions to APPROVED.
- **9. High-Impact Escalation**: Approve transfer > 2,000 -> requires safety confirmation checkbox.
- **10. Structured Rejection**: Reject with mandatory reason 'Clinical Need at Source Branch' and clinical note.
- **11. Quantity Override**: Override quantity from 50 to 30; observe live value recalculation.
- **12. Audit Trail**: Verify immutable audit records created for approval, rejection, and override.
- **13. Edge Cases Sandbox**: Verify all 8 failure prevention scenarios (transit > shelf life, closed branch).
- **14. Simulation Analytics**: Inspect 30-scenario Monte Carlo comparison (+7.44L gain, -52.09% loss).
- **15. Spatial Network**: View interactive topology map of 18 branches and Haversine distances.
- **16. RBAC Branch Scoping**: Sign in as 'Pharmacist 1' -> actions strictly scoped to PHARM-001 (403 on others).
- **17. Automated Tests**: Run `pytest backend/tests/test_backend.py` -> 49 passed, 0 failed in 50.03s.

7. Milestone Completion Status & Conclusion

PharmaShift has successfully implemented all Review #3 requirements and resolved all evaluator feedback from Review #2. The platform guarantees a secure **Login-First** user experience, isolates runtime rendering anomalies via **React Error Boundaries**, enforces rigorous **Role-Based Access Control**, validates all business logic with **49 passing automated unit tests across 16 categories**, and provides complete, transparent architectural documentation across the codebase.